

Course Name: INTRODUCTION TO STATISTICS

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

COMBINATORICS:- Permutation and Combination, Repetition and Constrained Repetition, Binomial Coefficients, Binomial Theorem.

UNIT-II

Frequency distributions, Histograms and frequency polygons, Measures of central tendency: Mean, Mode, Median, Dispersion, Mean deviation and standard deviation. Moments, Skewness, kurtosis.

UNIT-III

Elementary probability theory:- Definition, conditional probability, Probability distribution, mathematical expectation.

Theoretical distribution:- Binomial , poisson and Normal distribution, Relation between the binomial, poisoned Normal distribution.

UNIT-IV

Correlation and Regression:- Linear Correlation, Measure of Correlation, Least Square Regression lines.

Curve fitting:- Method of least square, least square line, least squares Parabola. chi-square test: definition of chi-square; signification test: contingency test, coefficient of contingency.

UNIT-V

Basic of sampling theory:- Sample mean and variance, students t-test, test of Hypotheses and significance, degree of freedom, Z-test, small and large sampling, Introduction to Monte Carlo method.